

**SUPSI**

University of Applied Sciences and Arts of Southern Switzerland

# Designing With

**A New Educational Module to Integrate  
Artificial Intelligence, Machine Learning and  
Data Visualization in Design Curricula**

**+**

**Didactic Guidelines**

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## 4. Module 2: Get In-depth

To face this module, students need some basic knowledge of the disciplines involved and some practice of tools. For this reason, the selection of tools provided to students within this module depends on the skills acquired during the first module or prior knowledge. To define the set of tools suitable, refer to the Designing With Framework [\[link\]](#). Working in groups (5-6 people) is recommended for this module, as collaboration facilitates addressing complex challenges. Additionally, a multidisciplinary approach incentivises different perspectives to emerge.

Students are first provided with a design challenge to drive the activity {MOD2-Act 1}. The design challenge provides a macro area for students to experiment with tools based on specific research questions and tasks. Examples of these challenges are *designing for climate change* and *designing for twin cities*. A detailed description is available from the Designing With Workshop repository [\[link\]](#).

To help students understand the challenge, case studies are provided. Working on challenges contributes to developing a knowledge dimension as students are asked to understand the interrelationship of basic elements (acquired in Module 1) and make them function together within a larger structure. Once the context is set, tools are introduced through the Designing With Framework {MOD2-Act 2}. The framework aims to introduce students to exploring tools of AI/ML/DV, understanding their capabilities, and applying them within a design process. Additionally, the framework promotes a data-based approach, bringing students closer to a design practice led by data input and output. The data-based approach is useful, especially in promoting the method of inquiry, which is part of the procedural knowledge dimension developed within this activity. Fostering the exploration of tools through the framework aims to infuse in students a method of design with AI.

During the process, students are asked to document this method. For this purpose, a structured template

[example] is introduced {MOD2-Act 3}. While in Module 1 documentation is one of the final activities, for the Module 2, keeping track of the steps, tools and choices at each of the design phases is encouraged to foster the acquisition of the method. During these activities, supervision from tutors and experts in the field is useful for choosing {MOD2-Act4} and properly applying AI tools {MOD2-Act5} for specific tasks defined based on the initial challenge. Two rounds of revisions are planned: the first one {MOD2-Act4} aims to support students in choosing the appropriate tools and refining the design concept; while the second one {MOD2-Act5} focuses on providing technical support in using the tools, as this second revision comes later when students are supposed to have the concept already defined and move to the prototyping and implementing phase. To allow teachers to assess the activities and future implementation of the module, students are asked for feedback. This activity contributes to creating awareness and knowledge of cognition. Lastly, to foster participation and sharing of results, students are asked to present their work and question others during a public presentation {MOD2-Act 7} (Fig. 10).

|                                   |                             |                   |              |                |                            |               |
|-----------------------------------|-----------------------------|-------------------|--------------|----------------|----------------------------|---------------|
| Module 2:<br>Get In-depth         | Cognitive process dimension |                   |              |                |                            |               |
|                                   | (1)<br>Remember             | (2)<br>Understand | (3)<br>Apply | (4)<br>Analyze | (5)<br>Evaluate            | (6)<br>Create |
| (a)<br>Factual<br>Knowledge       |                             |                   |              |                |                            |               |
| (b)<br>Conceptual<br>Knowledge    |                             | {MOD2-Act1}       |              |                |                            |               |
| (c)<br>Procedural<br>Knowledge    |                             |                   | {MOD2-Act2}  | {MOD2-Act3}    | {MOD2-Act4}<br>{MOD2-Act5} |               |
| (d)<br>Metacognitive<br>Knowledge |                             |                   |              | {MOD2-Act6}    | {MOD2-Act7}                | {MOD2-Obj}    |

Fig. 10 Module 2: 'Get In-depth'. Educational Objectives and Instructional Activities arranged into the Bloom Taxonomy Table.

Educational Objectives

|            |                        |                                 |                                               |
|------------|------------------------|---------------------------------|-----------------------------------------------|
| {MOD2-Obj} | Students will learn to | choose and apply AI/ML/DV tools | for specific activities of the design process |
|------------|------------------------|---------------------------------|-----------------------------------------------|

Instructional Activities

|         |                      |                    |                                                                  |
|---------|----------------------|--------------------|------------------------------------------------------------------|
| {Act.1} | Activity intended to | provide students   | with a design challenge                                          |
| {Act.2} | Activity intended to | support students   | in understanding, exploring and using the Framework              |
| {Act.3} | Activity intended to | provide students   | with a structured template for process and results documentation |
| {Act.4} | Activity intended to | supervise students | in choosing AI tools for specific design phases                  |
| {Act.5} | Activity intended to | supervise students | in applying AI tools for specific design phases                  |
| {Act.6} | Activity intended to | ask students       | feedback on the teaching module                                  |
| {Act.7} | Activity intended to | make students      | explaining and questioning the process and results               |

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